

A topological energy floor survives shrinking while the action cost vanishes

The classical 2+1 dimensional $O(3)$ sigma field has smooth degree-one static configurations of every radius, all with energy $4\pi\rho$. Their action magnitude over one radius-crossing time is $4\pi\rho R/c$, which tends to zero as the radius R shrinks. Thus a fixed topological sector can exclude zero energy without selecting a positive action floor. The same physical field has arbitrarily low-frequency vacuum waves.

Status: Q04 exploratory construction and written test; no accepted-ledger promotion or novelty claim. The sector bound and scale family are established sigma-model ingredients. The action and gap comparisons below are the task's use of them. This continuum classical field is a proposed physical model; it is not derived from the repository's Newtonian particle mechanics.

1. Field, action and admissible sector

Take a dimensionless unit vector $n(t, x) \in S^2$, $x \in \mathbb{R}^2$, with propagation speed $c > 0$ and stiffness $\rho > 0$ of energy units. Specify physical dynamics by the Lorentzian action

$$S[n] = \frac{\rho}{2} \int dt d^2x \left(c^{-2} |\partial_t n|^2 - |\nabla n|^2 \right). \quad (1)$$

Here S has energy-times-time units, and the canonical momentum density is $\pi = \rho \partial_t n / c^2$. The Hamiltonian is

$$H = \frac{\rho}{2} \int_{\mathbb{R}^2} \left(c^{-2} |\partial_t n|^2 + |\nabla n|^2 \right) d^2x. \quad (2)$$

Require finite energy and smooth extension of each spatial field to the one-point compactification, with $n(\infty) = e_3$. Its degree is

$$Q = \frac{1}{4\pi} \int n \cdot (\partial_1 n \times \partial_2 n) d^2x \in \mathbb{Z}. \quad (3)$$

Choose the sector $|Q| = 1$. Degree is conserved during smooth evolutions preserving the boundary extension. This is a restriction on admitted initial data and regular evolution; it does not explain why nature excludes $Q = 0$. No lattice spacing, lower radius or time window is imposed.

2. Square completion and an explicit shrinking family

Put $a = \partial_1 n$, $b = \partial_2 n$ and $q = n \cdot (a \times b)$. Tangency to the unit sphere gives

$$|a + n \times b|^2 = |a|^2 + |b|^2 - 2q, \quad |a - n \times b|^2 = |a|^2 + |b|^2 + 2q.$$

Integrating the appropriate square proves

$$H \geq E[n] := \frac{\rho}{2} \int |\nabla n|^2 d^2x \geq 4\pi\rho|Q|. \quad (4)$$

For every $R > 0$ set $r^2 = x_1^2 + x_2^2$ and

$$n_R(x) = \frac{(2Rx_1, 2Rx_2, r^2 - R^2)}{r^2 + R^2}. \quad (5)$$

Its norm is one, its boundary value is e_3 , and direct differentiation gives

$$|\nabla n_R|^2 = \frac{8R^2}{(r^2 + R^2)^2}, \quad q_R = -\frac{4R^2}{(r^2 + R^2)^2}.$$

Using $d^2x = 2\pi r dr$ and $\int_0^\infty r(r^2 + R^2)^{-2} dr = 1/(2R^2)$ yields $Q = -1$ and $E[n_R] = 4\pi\rho$. Spatial reflection gives the opposite degree. Equality in (4) makes each field a static energy minimizer in its sector, hence a stationary solution of (1). This constructs actual solutions, not only variational trial configurations.

The energy inside a disk of radius r is

$$E_R(r) = 4\pi\rho \frac{r^2}{r^2 + R^2}. \quad (6)$$

Consequently R is the half-energy radius, an intrinsic size rather than an arbitrary coordinate label. In the contraction $R \downarrow 0$, all members remain smooth and in the same sector. The limiting field is singular at the origin: away from it $n_R \rightarrow e_3$, while $n_R(0) = -e_3$. Their energy measures converge weakly to $4\pi\rho\delta_0$ and their gradients do not converge strongly to the vacuum in L^2 . Excluding the singular endpoint does not exclude the smooth sequence. This is a family of stationary preparations; no dynamical collapse or finite-inertia dilation mode is asserted.

3. Which action is being tested?

Define the candidate size-based action observable on this static family by

$$\mathcal{A}_R := E[n_R] \tau_R, \quad \tau_R := R/c.$$

For the static solution on a physical time interval of length τ_R , $S = -E\tau_R$, so this observable equals its action magnitude:

$$\mathcal{A}_R = |S[n_R; 0, R/c]| = \frac{4\pi\rho R}{c} \longrightarrow 0. \quad (7)$$

A crossing time measures propagation over the half-energy radius; it is not an oscillation period of this stationary field. Thus (7) tests this specified energy-time candidate, not every functional with action units. Equivalently, under spacetime dilation $n_\lambda(t, x) = n(t/\lambda, x/\lambda)$ on a time interval scaled by λ , (1) gives $S[n_\lambda] = \lambda S[n]$, whereas spatial degree is unchanged. The field equation retains this dilation symmetry at fixed ρ, c .

For a fixed externally prescribed duration $T > 0$, the same static configurations, each held for T , have $|S| = ET \geq 4\pi\rho T$. Its units and lower bound include that supplied clock. A canonical motion cost $\int dt \int \pi \cdot \partial_t n d^2x$ is exactly zero on every member of (5), even though its spatial gradient energy is positive. Topology excludes a spatially uniform field, but admits rest in configuration space.

The available dimensional parameters ρ and c cannot by themselves form an action: a length or a time must also enter. A lower radius $R \geq \ell > 0$ would give $\mathcal{A}_R \geq 4\pi\rho\ell/c$ within this family. That adds an independently justified physical scale and its coefficient; its survival as $\ell \downarrow 0$ would need a new mechanism. No physical-time attraction, preparation-independent constant or quantum phase rule follows from the stationary family.

4. The sector energy floor is not a frequency gap

The physical action (1), rather than an auxiliary sampling generator, determines small fluctuations. Near the uniform vacuum write $n = (u_1, u_2, \sqrt{1 - |u|^2})$. The quadratic action gives

$$\partial_t^2 u = c^2 \Delta u, \quad \omega(k) = c|k|. \quad (8)$$

The squared-frequency operator $-c^2\Delta$ on $L^2(\mathbb{R}^2; \mathbb{R}^2)$ with domain H^2 has spectrum $[0, \infty)$. For example normalized smooth packets $u_L(x) = L^{-1}f(x/L)$ have fixed L^2 norm and Rayleigh quotient $c^2 L^{-2} \|\nabla f\|_2^2 / \|f\|_2^2 \rightarrow 0$. On a periodic box of side L , after removing constant rotations, the first frequency is $2\pi c/L$; the thermodynamic limit closes it. These are vacuum linearized statements, not a quantized Hamiltonian spectrum.

Even within the degree-one sector there is no isolated static minimizer: translations $n_R(x-a)$ retain the same energy. Their tangent vectors $\partial_i n_R$ are square integrable by (4) and are zero directions of the static second variation. Quotienting translations is a further restriction; no positive internal fluctuation gap after such a quotient is claimed here. In particular (4) compares a chosen sector to the vacuum, not all excitations to a unique ground state within one sector.

5. Decision and next physical construction

Retain topology as a conservative mechanism for a positive *sector energy* at fixed stiffness. Park topology alone as an action selector in this model: it leaves an arbitrary radius, an admitted static state and low-frequency vacuum waves. This advances the zero-exclusion test beyond Q03's need for continuous power, while exposing separate size and motion requirements.

A concrete next candidate is a unit-vector field in three spatial dimensions with both quadratic and quartic spatial-gradient energies. Under a size rescaling their contributions behave as aR and b/R . Their competition can fix a radius without external fueling. The next decision is whether this stabilization supplies an action or only expresses a scale already in a, b and the kinetic normalization; it must also test the vacuum spectrum. Existence of a minimizer cannot be inferred from the one-parameter balance alone. This is the selected candidate test, not a new theorem or a claim that stabilization establishes quantum necessity.

Source and review

M. S. Ody and L. H. Ryder, *Time-Independent Solutions to the Two-Dimensional Non-Linear $O(3)$ Sigma Model and Surfaces of Constant Mean Curvature*, arXiv:hep-th/9402137v1 (1994), section 2, pp. 4–7, equations (2.1)–(2.26), supplies the static degree, energy and arbitrary-scale ingredients. Its unit stiffness gives $E = 4\pi|Q|$ for the duality solutions. The Lorentzian dynamics, the radius-crossing observable and the physical gap comparison are specified here.

The Q04 review records the written proof check and bounded primary-source comparison: one Sol-medium worker, two queries, one primary paper and six pages, with formula-image checks. The coordinator rechecked the square completion, radial integral, dilation and wave normalization, and visually checked source pp. 6–7. No numerical or symbolic verification scripts were used. Source matches concern the static ingredients; no novelty or accepted-ledger promotion is asserted.